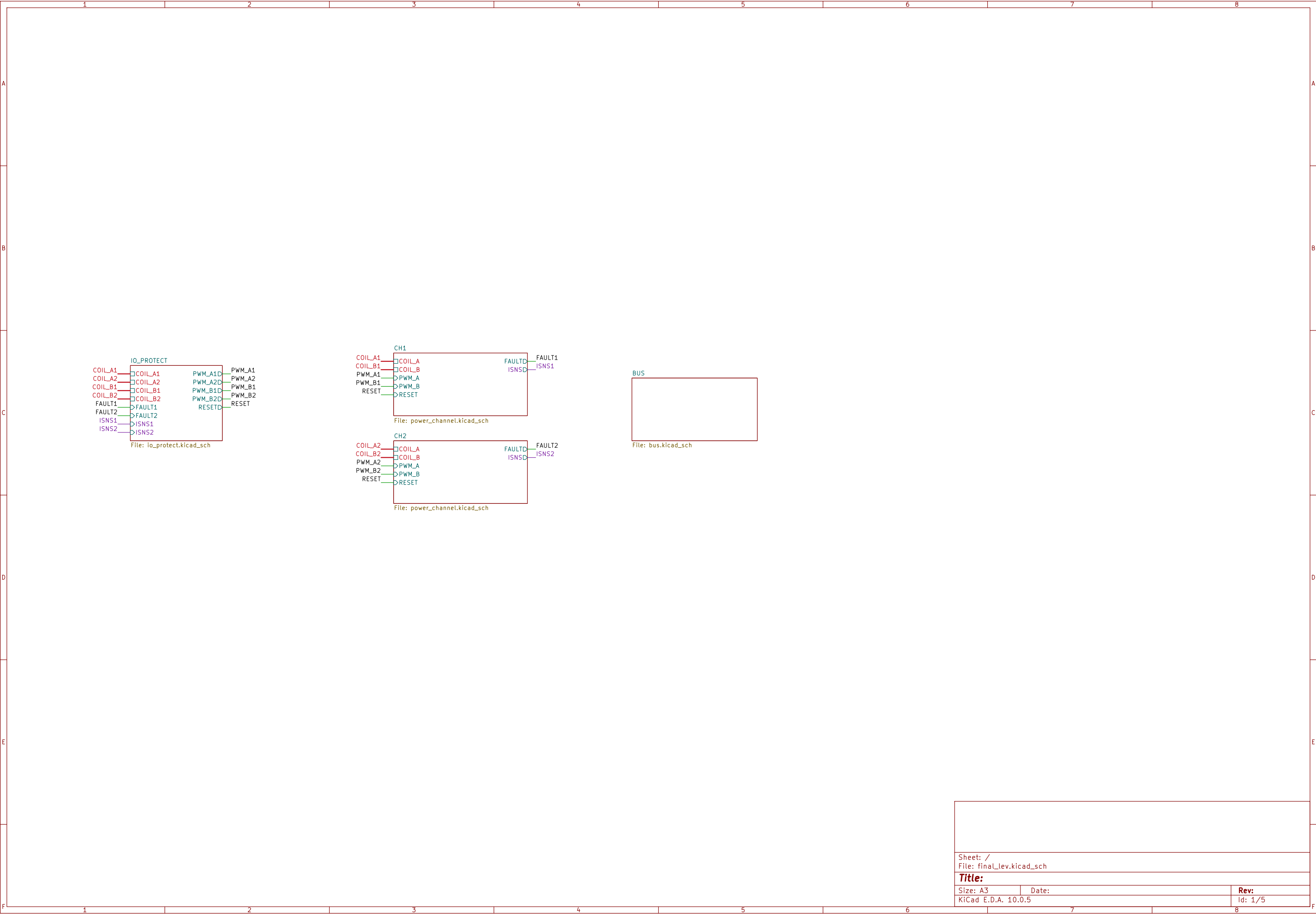
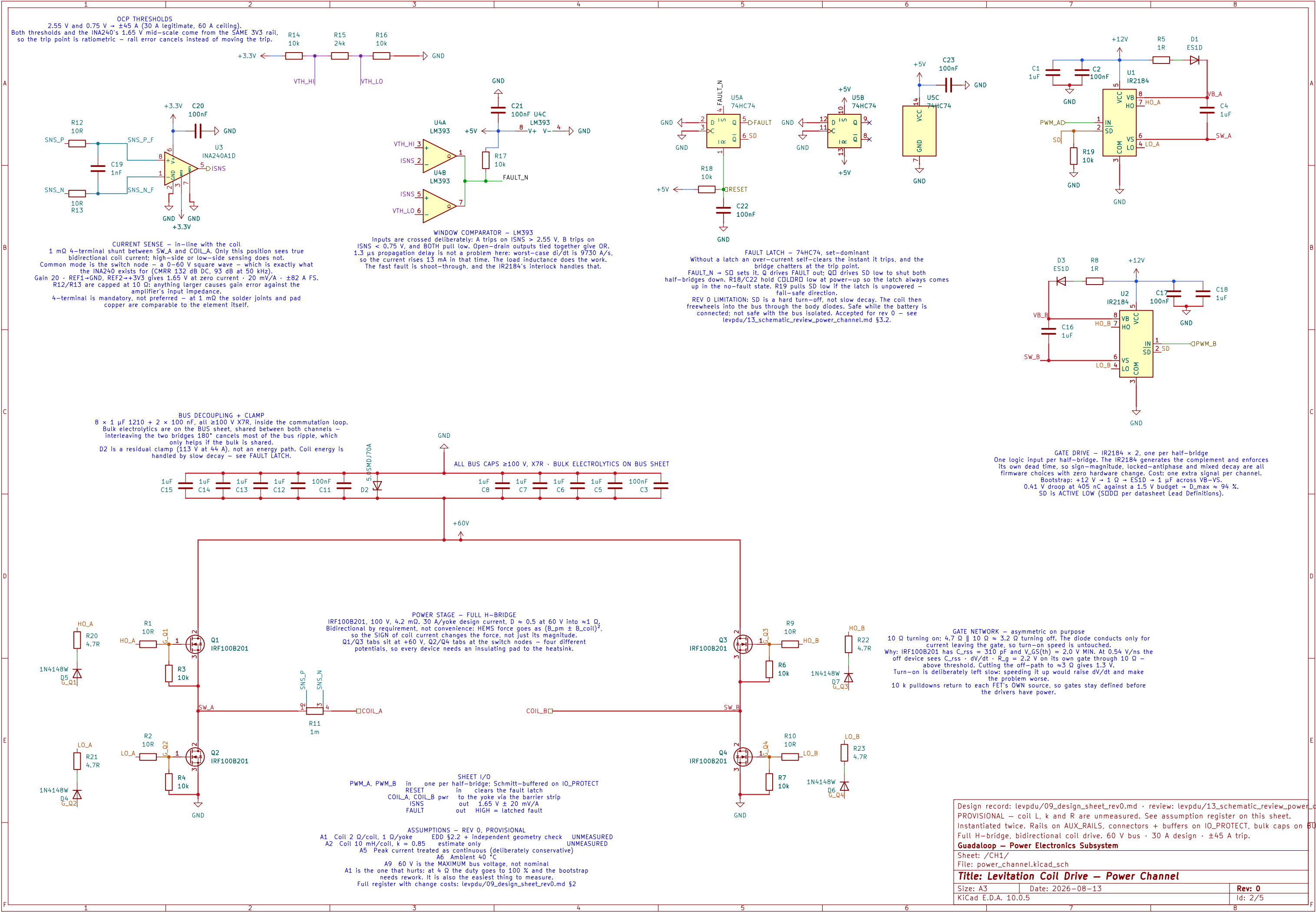


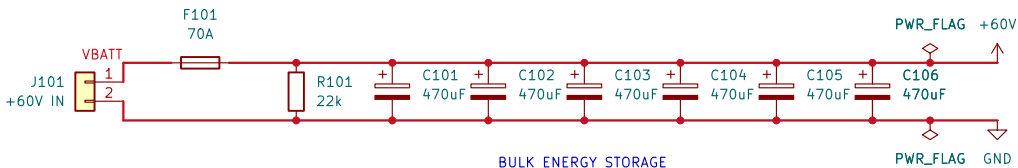




BUS ENTRY + FUSING

F101 protects everything downstream. It is NOT the over-current protection – that is the  $\pm 45$  A comparator on each power\_channel. This clears shorted-FET and shorted-bus faults, where current is limited only by battery impedance.

70 A: bus draw is coil current  $\times$  duty  $\approx$  30 A average per board with 60 A peaks. MIDI HP series is 70 VDC rated; most automotive fuses are 32 V and would be wrong here.



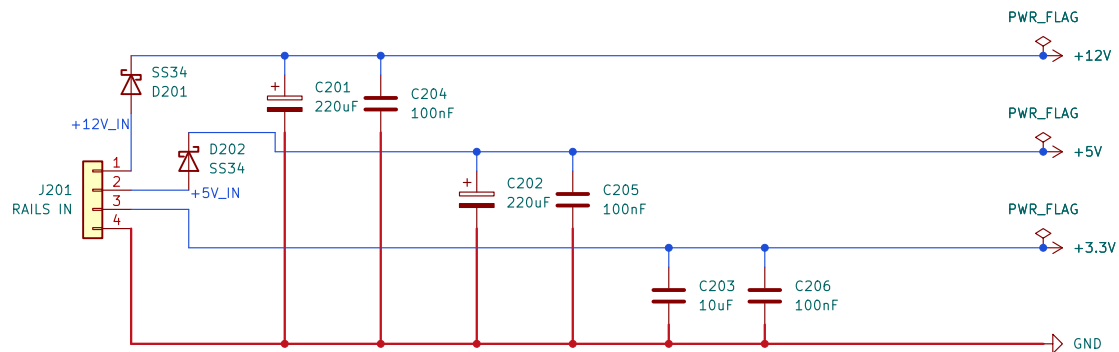
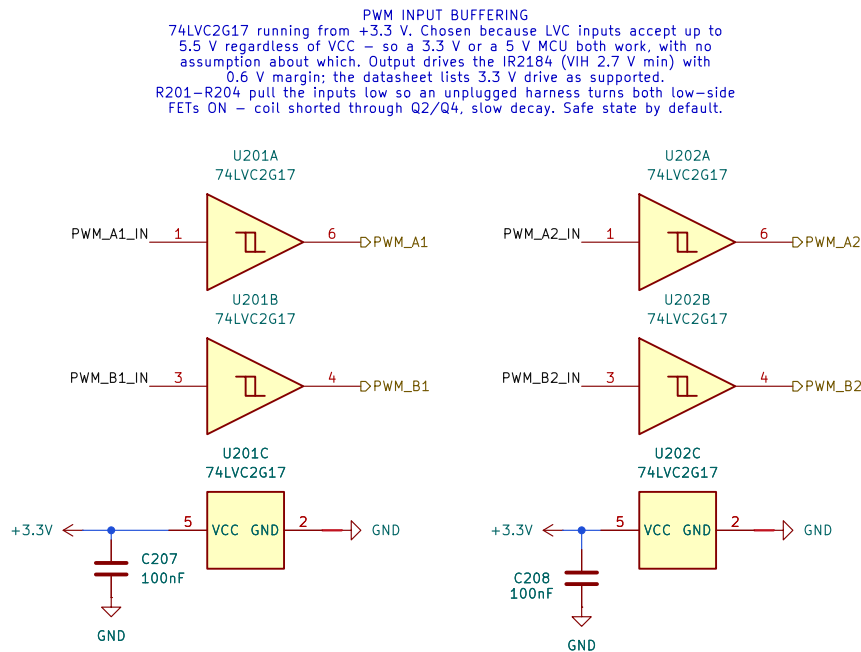
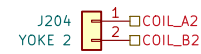
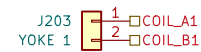
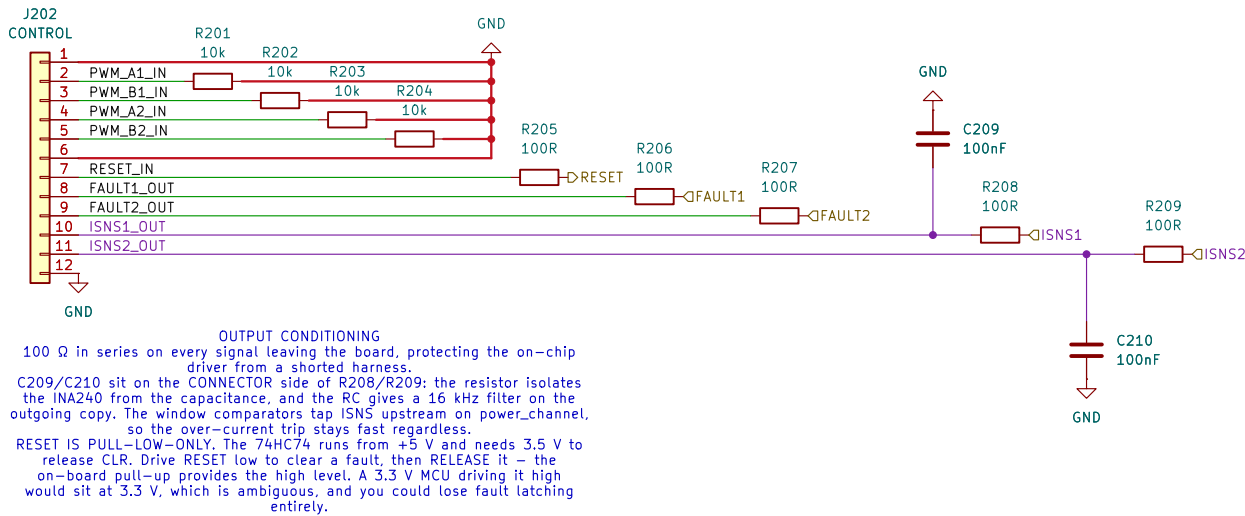
BULK ENERGY STORAGE

6  $\times$  470  $\mu$ F, shared by both power\_channel instances. Sized by RIPPLE CURRENT, not capacitance: ~20 A RMS on the shared bus once the two channels are interleaved 180°. Select parts so the ripple ratings sum past 20 A – the count follows the part.

R101 bleeds 5.1 J of stored energy: under 10 V in ~110 s.

Shared by both power\_channel instances.  
60 V battery entry, fusing, bulk energy storage, bleed.  
**Guadaloup – Power Electronics Subsystem**

|                                                         |                  |         |
|---------------------------------------------------------|------------------|---------|
| Sheet: /BUS/<br>File: bus.kicad_sch                     |                  |         |
| <b>Title: Levitation Coil Drive – Bus / Power Entry</b> |                  |         |
| Size: A4                                                | Date: 2026-08-13 | Rev: 0  |
| KiCad E.D.A. 10.0.5                                     |                  | Id: 3/5 |



**RAIL ENTRY**  
Rails are supplied by the pod – this board does not generate them. D201 does two jobs: blocks reverse polarity, and isolates C201 so the 220  $\mu$ F stays on this board when the pod's 12 V collapses. C201 is the hold-up for slow-decay-on-fault: the gate drivers must stay alive ~60 ms after the 60 V bus dies. 220  $\mu$ F over a 12~10.5 V window holds ~330  $\mu$ C, enough for ~5 mA of driver quiescent – and only because the gates are STATIC during slow decay. If anything is still switching, gate charge dominates and this is far too small.

Rails are supplied by the pod – this board does not generate them.  
Board I/O, PWM input buffering, rail entry and hold-up.

**Guadaloup – Power Electronics Subsystem**

Sheet: /IO\_PROTECT/  
File: io\_protect.kicad\_sch

**Title: Levitation Coil Drive – I/O and Protection**

|                     |                  |         |
|---------------------|------------------|---------|
| Size: A3            | Date: 2026-08-13 | Rev: 0  |
| KiCad E.D.A. 10.0.5 |                  | Id: 4/5 |

